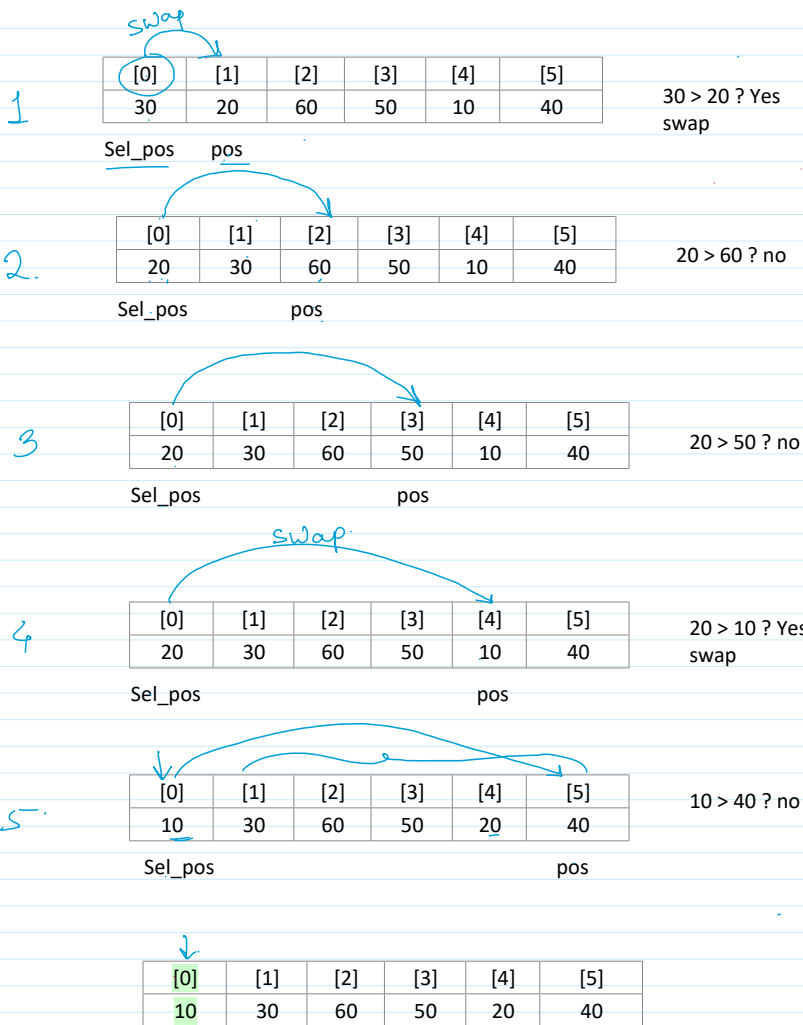
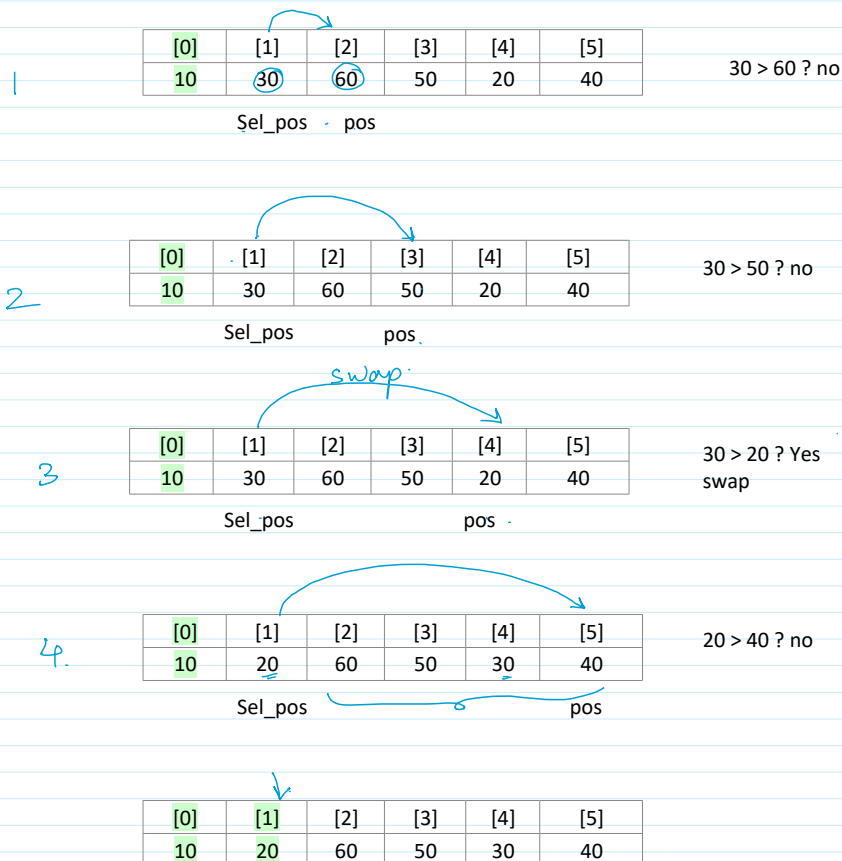


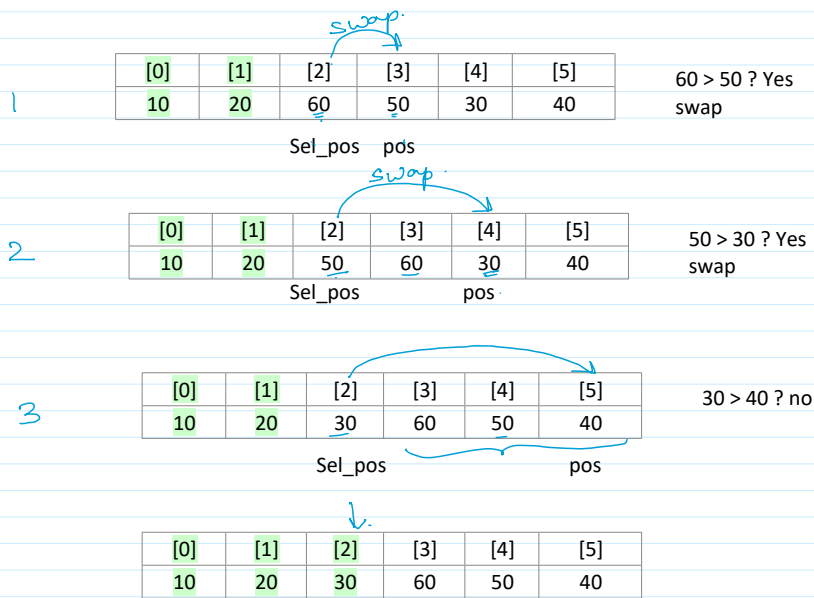
Iteration 1



Iteration : 2

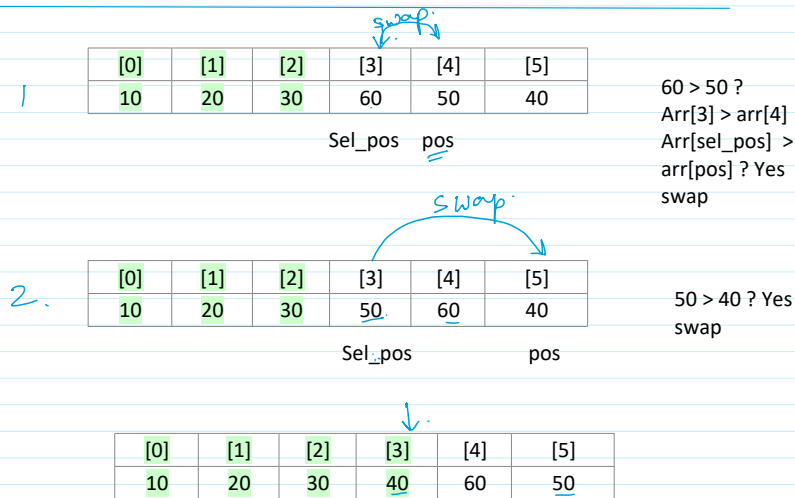


Iteration 3

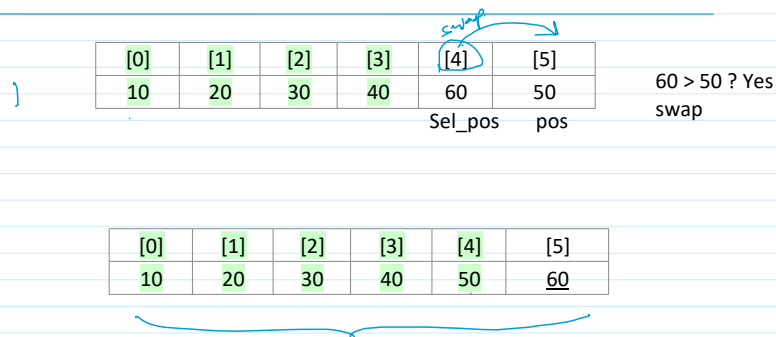


Iteration 4

cycle pass



Iteration 5



No. of iterations : $n-1 : 5$

No. of comparisons : $(n-1) + (n-2) + (n-3) \dots 1$

$$5+4+3+2+1 = 15$$

$6-1 = 5$ iterations

10 elements

9 iterations

$$9+8+7+6+5+4+3+2+1$$

Best Case Time Complexity : If the array is sorted : $O(n^2)$

Worst Case Time complexity : $O(n^2)$

Avg : $O(n^2)$

Space comp :
Input space : $O(n)$
Aux space : constant :
 $O(1)$

$\text{int arr}[\underline{3}][\underline{3}] \rightarrow O(n^2)$

```
for (i = 0; i < 3; i++)  
{  
    for (j = 0; j < 3; j++)  
    {  
        print  
    }  
}
```

$n \rightarrow 3$
inner $\rightarrow 9$.

$i = 0 \rightarrow j = 0$
 $j = 1$
 $j = 2$
 $j = 3 \times$ } 3

$i = 1 \rightarrow j = 0$ +
 $j = 1$ } 3
 $j = 2$
 $j = 3 \times$ +

$i = 2 \rightarrow j = 0$ +
 $j = 1$ } 3.
 $j = 2$
 $j = 3 \times$

$i = 3 \times$